

Key: The Finest Grain, Counted Twice

84-355 Democracy's Data | Georgia precinct returns | Instructor copy — do not distribute

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All code is given to students. Grade the interpretation.

Structure. The brief (`precincts-brief.Rmd`) walks ten numbered steps in prose with a D3 figure; the lab (`precincts.Rmd`) runs the same material with the code exposed and ends in four options. **Students should read the brief first.** Step numbers below refer to the brief.

This lab is the data spine of the course. Its parsed Georgia returns and the census-block shapefile in `data/` are read directly by `precinct-geography`, `levels-of-aggregation`, `areal-units` and `vote-dilution`, and `rpv` runs on the same county with the same registration logic. That changes how you should teach it: **an error here is not contained.** If a student mistrusts something in this session, they are entitled to mistrust four later ones, and the honest answer is that they are right to.

Every number below was computed from the built files at knit time.

The session in one sentence

Georgia's precinct returns are the finest-grained public record of American voting and the primary source for it — and the same state counted the same ballots a second time, got a different answer in 104 of 159 counties, and certified the second one, which means the finest data in the country is also the count that was superseded.

The findings

```
c(precincts      = nrow(p),
  counties       = nrow(co),
  presidential_votes = TOTV,
  margin_precinct_file = MARGIN,
  margin_recount    = RCMARGIN,
  counties_changed   = CHANGED,
  votes_moved        = MOVED,
  pct_election_day    = round(100 * mtot[[ED]] / sum(mtot), 1),
  mail_minus_eday_pts = round(mshare[[MB]] - mshare[[ED]], 1),
  largest_precinct    = max(p$total),
```

```
empty_precincts = sum(p$total == 0),
county_residual = MAXRESID)
```

```
#>      precincts      counties presidential_votes
#>      2656.0        159.0      4998482.0
#> margin_precinct_file margin_recount counties_changed
#>      12670.0      11779.0        104.0
#>      votes_moved    pct_election_day mail_minus_eday_pts
#>      2753.0         19.5         26.9
#> largest_precinct    empty_precincts county_residual
#>      15531.0         3.0         0.0
```

- **2,656 precincts in 159 counties, 4,998,482 presidential votes.** The complete state, from the state's own publication.
- **The precinct file says the margin was 12,670. The recount says 11,779.** The second is Georgia's certified number.
- **104 of 159 counties returned a different total the second time** — 65.4% of them — moving 2,753 votes in absolute terms, 0.055% of the ballots cast.
- **Only 19.5% of votes were cast in person on election day.** Mail and election-day electorates differ by 26.9 points of two-party share.
- **Precincts aggregate to counties with a residual of exactly 0** — 159 of 159 counties reproduced to the vote.

The three things to get said

1. Precision and authority are different properties. This is the most precise election data the United States produces, and it does not agree with the official result. Not because anyone erred — because “official” is a decision the state makes at a coarser resolution than the one it publishes. **Make somebody say the uncomfortable version out loud:** every racially polarized voting estimate in the second half of this course is built on the superseded count, because the recount was never published at precinct resolution. That is not a reason to distrust the analysis; it is a reason to know what it rests on.

2. A source that requires you to identify yourself is a source that can disappear. This lab used to run on VEST via a Harvard Dataverse guestbook, and on MEDSL before that. It now reads the Secretary of State directly. **The course now has no guestbooked sources at all, down from three.** The point is not that guestbooks are wrong — VEST's work is careful and the form is reasonable. The point is that a dependency you cannot script is a dependency somebody else controls, and going upstream past it bought something VEST does not carry: the vote-method column that Steps 5 and 6 are entirely about.

3. Step 7 is the session. Two counts, same ballots, same state, published as two archives. 2,753 votes moved and the outcome held. **Ask the room what that proves before you tell them.** Half will say it proves the count was accurate; half will say it proves hand and machine counts differ. Both readings use the same table, and both are defensible. The honest statement is the boring one — recounts move small numbers, and the margin was much larger than the movement.

The line for the board: the count is not a fact the data carries; it is a procedure the state performed, and Georgia performed two of them.

Walking the steps

Step	What it does	Min	Watch for
1 What one record is	County, precinct, registration, ballots, votes	5	The precinct is named, not coded. First sign this file came from 159 offices.
2 What the file contains	<code>structure.csv</code> , and the vote-method column	6	The lab's Part 2 goes further here — same file, three labels for one office.
3 Not a standard size	1 vote to 15,531	9	The biggest precincts are rural — both documents now say so. Take the guess first.
4 The state total	12,670-vote margin; exact aggregation	6	The clean aggregation is the setup for Steps 7 and 9. Say “hold onto that.”
5 The vote-method column	19.5% cast on election day	8	Students expect election day to dominate. It is fifth of the total.
6 Two electorates	26.9-point gap between mail and election day	12	The “red mirage” is arithmetic here, not allegation. Do that slowly.
— Predict, before Step 7	Counties changed, and by how much	3	Collect the guesses in writing. The step does not land without them.
7 Counted twice	104 of 159 counties; margin 12,670 to 11,779	14	The pivot. Nobody made a mistake; that is the hard part to convey.
8 Where the movement was	Fulton dominates the change	6	Largest movements in the largest counties is what <i>no</i> anomaly looks like.
9 The join that fails	Precincts and blocks do not nest	7	This is the bridge to precinct-geography and rpv . Name them.
10 What you can now say	Including what you cannot	4	End on the transferable finding, not on the recount.

That is 80 minutes with the prediction included.

Step 3 deserves the prediction too, and it is cheaper — ask before you show the table. Both student documents now print this evidence themselves (the `size-truth` chunk in the lab, the top-six table in the brief), so collect the guess first or the surprise is spent.

```
big <- head(np[order(-np$total), c("county", "precinct", "total")], 6)
big$one_precinct_county <- big$county %in% ONEPREC
print(big, row.names = FALSE)
```

```
#>   county                precinct total one_precinct_county
#> Lumpkin                Dahlonge 15531             TRUE
#> Coweta                 Newnan Centre 15134            FALSE
#> Stephens              Senior Center 11885            TRUE
#> Butts Butts County Admin Bldg 11771             TRUE
#> Jackson              Central Jackson 11757           FALSE
#> Jackson              West Jackson 11693           FALSE
```

```
c(largest_in_metro_five = METROMAX,
  one_precinct_counties = length(ONEPREC),
  votes_in_them         = sum(co$total[co$precincts == 1]))
```

```
#> largest_in_metro_five one_precinct_counties votes_in_them
#>                    5509                    16          87616
```

Ask where the largest precinct in Georgia is. The room will say Atlanta. It is **Dahlonge, in Lumpkin County, 15,531 votes** — and Lumpkin is one of the 16 counties that report the entire county as a single precinct. **The largest precinct anywhere in the metro five (Fulton, DeKalb, Gwinnett, Cobb, Clayton) is 5,509** — a third of the state record.

Be precise about what that does and does not show, because the seductive version is wrong. Metro precincts are *not* small: their median is 1,888 against 1,453 in the rest of the state. What metro counties do is **cap** them, because a county with 384 precincts is managing polling places. The extreme tail belongs to counties that never subdivided at all, so the maximum is a fact about administration and the median is a fact about population. **Two different things are being called “precinct size”**. Option C depends on the class having heard this.

If you are short on time, protect Steps 3, 6 and 7. The unit that is not a unit, the two electorates, and the count that was superseded are the argument. Steps 1, 8 and 10 can be read rather than worked.

Where this data comes from, and the two histories worth telling

The source moved upstream, past a gated one

Told as a sequence, this is the course’s provenance lesson in one lab. The lab ran first on MEDSL, then on VEST through a Harvard Dataverse guestbook — which returns, to a script, “You may not download this file without the required Guestbook response for guestbookID 458.” It now reads the Georgia Secretary of State directly: the XML county archives for 2020, and the 2024 JSON export at results.sos.ga.gov/cdn/results/Georgia/export-2024NovGen.json, parsed across all 159 counties by `data/parse-ga-sos.py`.

What forced the move was Cloudflare. `sos.ga.gov` returns 403 to a script and 200 to a browser on the identical URL with identical headers — verified, not assumed, and recorded in `data/README.md`. The irony is worth stating: the bot check is why the outer ZIP is still a manual download, and it is also why going upstream was worth doing at all, because it exposed how much of the course depended on somebody else’s willingness to keep serving a file.

Two things to be precise about in front of students, because both are easy to overstate:

- **Only the 2020 route is gated.** The 2024 JSON is on a CDN path that is *not* behind the bot check and is script-fetchable (`--json-url` fetches it unattended). So the lab’s “everything is gated” framing is true of the archive and false of the newer platform. Georgia is in the middle of fixing this without saying so.

- **The route does not generalise.** It exists because one state publishes structured county exports. Forty-nine others do something else, which is why VEST existed in the first place and why the volunteer project is not obsolete.

The parser had to be taught what the race is called

```
data.frame(
  source      = c("XML archives (2012 - June 2024)", "JSON export (Nov 2024)"),
  contest_label = c(getv(s20, "contest"), getv(s24, "contest")),
  precincts    = c(getv(s20, "precincts"), getv(s24, "precincts")),
  candidates    = c(getv(s20, "candidates"), getv(s24, "candidates")),
  stringsAsFactors = FALSE)
```

Source	Contest label	Precincts	Candidates
XML archives (2012 - June 2024)	President of the United States	2656	3
JSON export (Nov 2024)	President of the US	2701	6

```
cat(paste0("2020 vote-method labels: ", getv(s20, "vote methods"), "\n",
          "2024 vote-method labels: ", getv(s24, "vote methods"), "\n"))
```

```
#> 2020 vote-method labels: Absentee by Mail Votes | Advanced Voting Votes | Election Day Votes | Provi
```

```
#> 2024 vote-method labels: (not reported) | Absentee by Mail | Advanced Voting | Election Day | Provis
```

The same state, the same race, two labels. The XML calls it *President of the United States*; the 2024 JSON calls it *President of the US*. The vote-method labels drift too — **Election Day Votes** becomes **Election Day**. The parser's default contest pattern is the longest prefix common to both.

This is the best concrete argument in the course against joining on human-readable names, and it is better than the abstract version because the failure it prevents is documented in the script. An earlier pattern matched the bare substring **President** and swept in **Co Commission Chair/Presidente de la Comisión del Condado** — County Commission Chair, in Spanish, from Georgia's bilingual counties — putting a Gwinnett commission candidate into the presidential totals with 233,110 votes. **A join key that is a sentence in a natural language is a join key that changes when somebody edits a label.**

One correction to make if you tell this story. The parser handles a candidate string carrying more than one parenthetical tag by reading the *last* one, and strips (I) first because in this source (I) means **incumbent, not Independent**. That is a real repair and worth a sentence. But **the scripts in this folder do not contain fusion-voting handling or footnote repair** — Georgia does not permit fusion voting, and no footnote markers appear in these exports. If you have heard those two attached to this lab, they belong to a different state's parser. Do not repeat them; a key that oversells its own provenance story teaches the opposite of the lesson.

The 489,876-vote join, and whether this lab would have caught it

Tell this story here, not only in precinct-geography. The sibling lab's first crosswalk attached votes to the 2020 shapefile by matching precinct names. It reached 90.4% and lost **489,876 votes, 9.8% of**

Georgia, concentrated in Chatham and DeKalb. Two rounds of name normalisation moved the loss from 11.1% to 9.8% and then not at all. The shapefile already carried the vote totals in its own attribute table; there was never a join to make.

(That figure is history, recorded in the header of `data/build-crosswalk.R`. It is not recomputable from any current file, and it is the only number in this key that is typed rather than computed. Flag it as such if a student asks.)

Would this lab's checks have caught it? Half yes, and the half matters.

- **The automated checks would not have.** They are internal to the parse: `parse-ga-sos.py` reconciles each candidate's stated total against the sum of their precinct-by-method rows, and `build-data.R` asserts `nrow(co) == 159`. Both would pass while an entirely separate join to a shapefile threw away a tenth of the state. **A check inside one file cannot see a mistake made between two files**, and that is the general form of the lesson.
- **The habit the lab teaches would have caught it instantly.** Step 4's whole purpose is to sum the parts and compare against an independently published total:

```
c(counties_reproduced_exactly = EXACT, of = nrow(mg), worst_residual = MAXRESID)
```

```
#> counties_reproduced_exactly      of
#>                        159      159
#>      worst_residual
#>                        0
```

A residual of 0 across 159 counties is what a correct aggregation looks like. Nobody ran that comparison on the name join. Had they totalled the joined file once and put it next to 4,998,482, the 489,876 would have announced itself in one line.

So the answer to give the class is: the check that would have caught it is the one this lab performs in Step 4 and the sibling lab did not perform until after the damage — and the reason it went unnoticed is that the joined file *looked fine*, which is what 90.4% always looks like.

The carry that shipped loses 22 votes of 13,821,357, a share of 0.000%.

Before you teach this: one defect, and two repairs

Written down so you are not surprised in the room. Two of the three problems this section used to list are fixed in the student documents; the third is in a script and is still live.

Repair one: the lab's Part 2. It used to filter `structure.csv` on a column named `kind` that a rebuild no longer produces. It did not error — it printed `vote_columns 0` and `<0 rows>` — and then taught VEST's G2OPRETRU convention, a file this lab has not read since the move to the Secretary of State. Part 2 now reads the real `structure.csv` (7 rows, columns item, value) and teaches **the contest-name and vote-method drift from the section above**, computed at knit time from `ga2020_structure.csv`, `ga2020_vote_methods.csv` and `ga2024_structure.csv`, with the Gwinnett trap printed straight out of the header of `data/parse-ga-sos.py`. G2OPRETRU survives in the lab as an explicitly labelled historical contrast: two designs, two failure modes.

So the lab and this key now overlap on purpose. Students meet the label drift in Part 2; you have the same material here plus the parenthetical corrections. **Do not teach it twice** — in the room, go straight

to the question the lab sets up and this key answers: a code cannot drift but cannot explain itself, a sentence explains itself but drifts, and *both* files are unusable by somebody who was not told.

Repair two: the largest-precinct sentence. Both documents used to follow 15,531 with “the word means one thing in rural Georgia and something else in metropolitan Atlanta,” which students hear as *metro precincts are big* — the opposite of what the tail shows. Both now lead with the rural tail and the metropolitan cap, computed at knit time, and land on the version this key has always given: **the word means one thing in a county with 384 precincts and another in a county with one.** The lab prints the top six and the metro comparison in its `size-truth` chunk; the brief prints the same six as a table before its county spread. **The correction is no longer yours to make in the room** — what is still yours is the prediction, and Option C still depends on the class having heard the distinction out loud.

The defect that is still live: the 2024 file is 18 votes short at precinct level, and the script says it is not. The header of `data/parse-ga-sos.py` reports that 84 rows have precinct sums falling short of the candidate total and states “**No statewide contest is affected.**” That is wrong. Summing the 2024 JSON’s presidential precinct rows gives Trump 2,663,110 and Harris 2,548,006 against county-level totals of 2,663,117 and 2,548,017 — **7 and 11 votes, all in Fulton County.** It changes nothing substantive (18 votes in 5.25 million) and it matters anyway, because the claim in the script is exactly the kind of reassurance the lab teaches students not to accept unverified. **If you use the 2024 file, say the number.**

Option A — The empty precincts

Verified output

```
p[p$total == 0, c("county", "precinct", "registered", "ballots_cast",
                 "total", "dem_two_party_pct")]
```

County	Precinct	Registered	Ballots cast	Total	Dem two party pct
Fulton	CP084	1	0	0	NA
Fulton	SC08H	1	0	0	NA
Fulton	UC01D	6	0	0	NA

```
c(empty_precincts = sum(p$total == 0),
  na_dem_share    = sum(is.na(p$dem_two_party_pct)),
  mean_dem_share_naive = mean(p$dem_two_party_pct),
  mean_dem_share_narm  = round(mean(p$dem_two_party_pct, na.rm = TRUE), 2),
  statewide_two_party  = round(100 * sum(p[[DEM]]) / sum(p[[DEM]] + p[[REP]]), 2))
```

```
#>      empty_precincts      na_dem_share mean_dem_share_naive
#>              3.00              3.00              NA
#> mean_dem_share_narm statewide_two_party
#>              50.64              50.13
```

3/4: Finds all 3, reports that they are in Fulton County with almost no registered voters and no ballots cast, and concludes they are administrative rather than real.

4/4: Notices they carry a registration count but zero ballots, works out that dropping them cannot change a single vote total, and identifies what they *do* change — Fulton’s precinct count and the state’s, and therefore every per-precinct average.

4/4, best: Uses them to catch a live bug in the file’s own design. The naive `mean(dem_two_party_pct)` is **NA** because a zero-vote precinct has an undefined share; the `na.rm = TRUE` version is 50.64, and the statewide two-party share is 50.13. **Three different answers to “how Democratic was Georgia”, and the gap between the last two has nothing to do with the empty precincts** — it is the difference between averaging precincts and averaging voters, which is the same “per what?” problem Step 3 raised about size. A student who gets from three blank rows to that distinction has done the best available work on this option.

Option B — Aggregate up

Verified output

```
mismatch <- mg[mg$t != mg$total, c("county", "t", "total")]
c(counties      = nrow(mg),
  reproduced_exactly = EXACT,
  mismatched       = nrow(mismatch),
  worst_residual   = MAXRESID,
  precinct_counts_match = sum(as.vector(table(p$county)) ==
                              co$precincts[order(co$county)]))
```

```
#>           counties reproduced_exactly mismatched
#>           159             159             0
#> worst_residual precinct_counts_match
#>           0             159
```

```
head(mg[order(-mg$total), c("county", "precincts", "t", "total")], 5)
```

County	Precincts	T	Total
Fulton	384	524659	524659
Gwinnett	156	413865	413865
Cobb	145	393746	393746
DeKalb	191	370804	370804
Cherokee	42	144830	144830

3/4: Confirms the aggregation is exact — 159 of 159 counties, residual 0 — and says so.

4/4: Answers the harder half. Precincts nest inside counties **because the county drew them inside its own border**; blocks and precincts do not nest because two different authorities drew them at two different times for two different purposes, and neither is required to respect the other. The distinction is authorship, not geometry.

4/4, best: Notices that the exactness is weaker evidence than it looks. Both files were produced by the same parser from the same source, so the check verifies the arithmetic and nothing about the underlying count — it is **internal consistency, not validation**. The real external check in this lab is Step 7’s recount, and it *fails*: the same election, independently recounted, disagrees. A student who says “Step 4 passing does not license trusting Step 4’s totals, and Step 7 proves it” has understood something the brief only gestures at.

Option C — Size and politics

Verified output

```
q <- cut(np$total, quantile(np$total, 0:5 / 5), include.lowest = TRUE,
        labels = paste0("Q", 1:5))
wt <- tapply(seq_len(nrow(np)), q,
            function(i) 100 * sum(np[[DEM]][i]) / sum(np[[DEM]][i] + np[[REP]][i]))
data.frame(quintile = levels(q),
           precincts = as.vector(table(q)),
           median_size = as.vector(round(tapply(np$total, q, median))),
           mean_precinct_dem = round(as.vector(tapply(np$dem_two_party_pct, q, mean)), 1),
           vote_weighted_dem = round(as.vector(wt), 1))
```

Quintile	Precincts	Median size	Mean precinct dem	Vote weighted dem
Q1	532	428	46.7	45.2
Q2	529	1017	47.9	48.1
Q3	533	1630	55.5	55.4
Q4	528	2324	54.1	54.1
Q5	531	3537	49.1	46.7

```
c(correlation_size_vs_dem_share =
  round(cor(np$total, np$dem_two_party_pct, use = "complete.obs"), 3))
```

```
#> correlation_size_vs_dem_share
#> -0.025
```

```
# who is actually in the two tails
sapply(c("Q1", "Q5"), function(k) names(head(sort(table(np$county[q == k]),
                                                decreasing = TRUE), 4)))
```

```
#>      Q1      Q5
#> [1,] "Fulton"  "Gwinnett"
#> [2,] "Richmond" "Cobb"
#> [3,] "Chatham" "Fulton"
#> [4,] "Dodge"   "Cherokee"
```

3/4: Plots it, reports that the correlation is -0.025 — near zero — and concludes there is no relationship between precinct size and vote share.

4/4: Notices that “no correlation” is the wrong summary because the pattern is not linear. The middle quintiles are the most Democratic and both tails are less so; a single correlation coefficient averages an arch down to nothing. Says what a scatterplot shows that `cor()` hides.

4/4, best: Says what the variable is actually measuring. Sorting precincts by size mostly sorts **counties**: the smallest quintile is 23.9% Fulton — the county that subdivides most finely — mixed with rural south Georgia, and the largest quintile is the metro ring, Gwinnett and Cobb and Cherokee. **Fulton appears in both tails**, which is the tell: a county with 384 precincts contains the whole range by itself, so the variable is not sorting places, it is sorting *within and across* counties at the same time. **The arch is a composition effect**. It runs from urban-plus-rural, through the Democratic inner suburbs, out to the less Democratic

outer ones, and precinct size is the incidental variable that happens to order them that way. **The apparent finding is a fact about who lives where and how counties draw polling places, wearing the costume of a fact about precinct size.** Anyone who reaches that, and connects it to Step 3’s warning that the unit is not comparable across counties, has the best answer available. Reward equally a student who declines to interpret it at all on those grounds.

One thing to correct if a student asserts it: the *largest single precincts* are rural (Step 3), but the *largest quintile* is suburban. Only 2.1% of precincts in Q5 sit in a one-precinct county. Both facts are in the data and they are about different parts of the distribution.

Option D — Your own question

No fully verified output; grade the reasoning.

3/4: Asks a question the file can answer and answers it correctly.

4/4: States in advance what the data would have to look like for the answer to be different, then checks. Any question about vote method, roll-off between ballots cast and presidential votes, or 2020-vs-2024 county movement is well served by these files.

4/4, best: Picks a question the file *cannot* answer and demonstrates why from the file itself — no boundaries, no demographics, no individual ballots, one state, one election, and the count that was superseded. **Establishing a limit empirically is worth more than answering an easy question**, and this is the one option where a student can do that without leaving the folder.

Grading

Four-point scale from the syllabus.

- **Never penalise code — it is given.** Every chunk in the lab runs as distributed; a student who reports what came out has done the assigned work.
- **The move that earns a 4** is treating a property of the data as a consequence of who produced it and how. Precinct size is county practice; the method split is administrative channel; the recount gap is procedure.
- **A 2** is “the data is messy so we cannot trust it.” True, useless, and the thing the lab exists to move past.
- **Also a 2:** quoting the 12,670 margin as *the* result after Step 7 has explained that the certified figure is 11,779.
- **Reward anyone who asks which count the downstream labs use.** The answer is the precinct file, because it is the only one published at that resolution, and noticing the question is more sophisticated than most of the available answers.

Anticipated questions

- “*So which count is right?*” — The recount is certified, so legally the recount. Neither is “right” in the sense the question means. A recount is a second performance of a procedure, not an appeal to a ground truth, and the ballots are the only ground truth there is.

- “*Why did 65% of counties change?*” — Because hand-counting five million ballots produces different handling errors than machine-counting them. 2,753 votes across 4,998,482 is 0.055%. That rate is the finding; the direction is not.
- “*Isn’t 19.5% on election day suspicious?*” — No, and this is the question to prepare for. 2020 was a pandemic election with three weeks of in-person early voting in Georgia. **Advanced Voting alone is 53.9% of the total.** The file shows the split without any claim attached to it, which is precisely why it is worth having.
- “*Could we do this for another state?*” — Not this way. The route exists because Georgia publishes structured county exports. That is the honest answer and it is also the argument for why VEST existed.
- “*Why does the state publish 159 files instead of one?*” — Because the counties are the authority and the state is compiling, not counting. The same fact drives **precinct-geography**’s Step 1.

Connections

- **precinct-geography** — runs on this folder’s shapefiles and crosswalks and is the direct sequel. Teach this one first; that lab assumes the returns are understood.
- **levels-of-aggregation** — reads `ga2020_precincts.csv`, `ga2020_vote_methods.csv` and the 2024 files directly. The same votes at four resolutions.
- **rpv** and **vote-dilution** — both need returns and demographics in the same place, which is Step 9’s unsolved join. **vote-dilution** reads the census blocks that live in this folder.
- **areal-units** — shares the block shapefile and generalises Step 9 into the modifiable areal unit problem.
- **residual-votes** — the roll-off visible here between ballots cast and presidential votes (28,202 ballots, 0.56%) is that lab’s whole subject, measured at this lab’s resolution.
- **data-sources** — the provenance argument in miniature. This is the lab where the course stopped depending on a form.
- **election-night** — the red mirage in Step 6 is the arithmetic behind that session’s reporting sequence. They pair well in a single board post.